

Nice! You're really getting this stuff.

Your time: 6 mins



Correct

19

Incorrect

1

Next steps

Take a new test



Try another test to boost your confidence.

Review with Flashcards



Study each term as a flashcard to improve recall.

Your answers

1 of 20

Term 

Diagram of the "pyramidal" and "extra-pyramidal" motor tracts of the CNS

You're doing great!

Review flashcards

New test

Don't know?

2 of 20

Term 🔊

What are the 4 structures of the Pons?

Nice work!

1. Autonomic nerve pathways
2. Integration of information
3. **Exerts some control over:**
 - regulation of respiration
 - involuntary actions
 - sensory roles: hearing, equilibrium, taste, facial sensations (touch & pain)
 - motor roles: eye movement, facial expressions, chewing, swallowing, secretion of tears/saliva

1. Hypoglossal nucleus
2. Solitary tract
3. Trigeminal nerve
4. Olivary nucleus
5. Pyramidal tract
6. Nucleus ambiguus

2. Pontine tegmentum (dorsal pons)

- 3. Lemniscus = secondary sensory fiber
- 4. Fasciculus = nerve and motor bundles

1. Autonomic Control

- Cardiac = sympathetic and parasympathetic functioning
- Respiratory = chemoreceptors (detect blood-oxygen concentration)
- Reflexes = vomiting, coughing, sneezing, swallowing
- Vasomotor = baroreceptors (detect blood-pressure changes)

2. Decussation of sensory fibers

- Obex

Don't know?

3 of 20

Term 🔊

Diagram for the cross-section brainstem

Awesome!

Term 🔊

Outline the "pyramidal" motor tracts of the CNS

Excellent!

2 major divisions:

1. Basilar part of the pons (ventral pons)

2. Pontine tegmentum (dorsal pons)

3. Lemniscus = secondary sensory fiber

4. Fasciculus = nerve and motor bundles

1. Hypoglossal nucleus

2. Solitary tract

3. Trigeminal nerve

4. Olivary nucleus

5. Pyramidal tract

6. Nucleus ambiguus

1. Corticospinal tract (voluntary control of body)

- Lateral corticospinal

- Anterior corticospinal

2. Corticobulbar tract (voluntary control of face, head, and neck)

1. Rubrospinal (fine motor control)

2. Reticulospinal

- Medial reticulospinal (contraction increases tone)

- Lateral reticulospinal

4. Tectospinal/Colliculospinal (head co-ordination)

Don't know?

5 of 20

Term 🔊

What are the functions of the Medulla?

You've got this!

1. Hypoglossal nucleus
2. Solitary tract
3. Trigeminal nerve
4. Olivary nucleus
5. Pyramidal tract
6. Nucleus ambiguus

1. Autonomic Control

- Cardiac = sympathetic and parasympathetic functioning
- Respiratory = chemoreceptors (detect blood-oxygen concentration)
- Reflexes = vomiting, coughing, sneezing, swallowing
- Vasomotor = baroreceptors (detect blood-pressure changes)

2. Decussation of sensory fibers

- Obex

1. Autonomic nerve pathways
2. Integration of information
3. **Exerts some control over:**

- motor roles: eye movement, facial expressions, chewing, swallowing, secretion of tears/saliva

2 major divisions:

1. Basilar part of the pons (ventral pons)

2. Pontine tegmentum (dorsal pons)

3. Lemniscus = secondary sensory fiber

4. Fasciculus = nerve and motor bundles

Don't know?

6 of 20

Term 🔊

What are the 12 cranial nerves (CN's)?

You're doing great!

1. CN-I = Olfactory (smell)
2. CN-II = Optic (vision)
3. CN-III = Oculomotor (eyelid opening and pupil constriction)
4. CN-IV = Trochlear (vertical/inward eye-movement)
5. CN-V = Trigeminal (sensory from nose, mouth, & scalp) (motor for chewing)
6. CN-VI = Abducens (lateral/outward eye-movement)
7. CN-VII = Facial (sensory for taste from anterior of tongue) (motor for facial expressions)
8. CN-VIII = Vestibulocochlear (senses of hearing and balance)
9. CN-IX = Glossopharyngeal (sensory for taste from posterior of tongue) (motor for swallowing)
10. CN-X = Vagus (sensory and motor as its function is the autonomic regulation of viscera: heart, lungs, digestive tract)
11. CN-XI = Accessory (motor for neck/shoulders to control head-movement)
12. CN-XII = Hypoglossal (motor for tongue to control speech and swallowing)

1. Autonomic nerve pathways
2. Integration of information
3. **Exerts some control over:**
 - regulation of respiration
 - involuntary actions
 - sensory roles: hearing, equilibrium, taste, facial sensations (touch & pain)
 - motor roles: eye movement, facial expressions, chewing, swallowing, secretion of tears/saliva

1. Hypoglossal nucleus
2. Solitary tract
3. Trigeminal nerve
4. Olivary nucleus
5. Pyramidal tract
6. Nucleus ambiguus

1. Autonomic Control

- Reflexes = vomiting, coughing, sneezing, swallowing
 - Vasomotor = baroreceptors (detect blood-pressure changes)
2. Decussation of sensory fibers
- Obex

Don't know?

7 of 20

Term 🔊

What are the 6 structures of the Medulla?

Nice work!

1. Autonomic nerve pathways
2. Integration of information
3. **Exerts some control over:**
 - regulation of respiration
 - involuntary actions
 - sensory roles: hearing, equilibrium, taste, facial sensations (touch & pain)
 - motor roles: eye movement, facial expressions, chewing, swallowing, secretion of tears/saliva

1. Hypoglossal nucleus
2. Solitary tract
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1. Autonomic Control

- Cardiac = sympathetic and parasympathetic functioning
- Respiratory = chemoreceptors (detect blood-oxygen concentration)
- Reflexes = vomiting, coughing, sneezing, swallowing
- Vasomotor = baroreceptors (detect blood-pressure changes)

2. Decussation of sensory fibers

- Obex

2 major divisions:**1. Basilar part of the pons (ventral pons)****2. Pontine tegmentum (dorsal pons)**

3. Lemniscus = secondary sensory fiber

4. Fasciculus = nerve and motor bundles

Don't know?

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Definition 🔊

1. Autonomic nerve pathways

2. Integration of information

3. Exerts some control over:

- regulation of respiration
- involuntary actions
- sensory roles: hearing, equilibrium, taste, facial sensations (touch & pain)
- motor roles: eye movement, facial expressions, chewing, swallowing, secretion of tears/saliva

Awesome!

What are the functions of the Medulla?

What are the 12 cranial nerves (CN's)?

What are the functions of the Pons?

What are the 4 structures of the Pons?

Don't know?

9 of 20

Term 🔊

What is the "Hypoglossal nucleus?"

Not quite, you're still learning!

cranial nerve v:
-face sensations
-biting and chewing motor functions

✓ cranial motor nerve nucleus

✗ cranial nerves ix and x:
-speech
-swallowing

stretch and chemoreceptors of cardiovascular, respiratory, and intestinal tracts

Don't know?

10 of 20

Definition

- UMN's originate from multiple places in the motor cortex in the cerebral cortex
- The neurons descend as the **corticospinal tract** down to the midbrain.
- Continues down through the cerebral peduncles (in the midbrain), the basis pontis (in the pons), and on through the pyramidal tract (in the medulla)
- Immediately after passing the medulla, 1 of the neurons continues straight, and it ends at the anterior "uncrossed" corticospinal tract (LMN in the Cervical cord).
- The other 3 neurons then branch off at the **pyramidal decussation** ending at either the lateral "crossed" corticospinal tract or the spinal motor neurons, in either the Cervical cord or the Lumbar cord LMN

OVERVIEW: all these nuclei contain neurons that allow for voluntary motor control of limbs and voluntary posture adjustment

Excellent!

Dive deeper and explain the pathways for the "Corticospinal" pyramidal tract (in the context of uploaded diagram)

Dive deeper and explain the pathways for the "Corticobulbar" pyramidal tract (in the context of uploaded diagram)

Dive deeper and explain the pathways for the "Vestibulospinal" extra-pyramidal tract (in the context of uploaded diagram)

Dive deeper and explain the pathways for the "Rubrospinal" extra-pyramidal tract (in the context of uploaded diagram)

Don't know?

11 of 20

Definition

- UMN's of the cranial nerves originate from the precentral gyrus in the cerebral cortex
- The neurons descend as the **corticobulbar tract** down to the midbrain. Neurons branch off to synapse with 3 nuclei: CN-III, CN-IV, and CN-VI
- Continues down through the cerebral peduncles (in the midbrain) to the pons. Neurons branch off to synapse with 2 nuclei (LMN's): CN-V and CN-VII
- It ends, continuing down through the basis pontis (in the pons) to the medulla oblongata (in the medulla). Neurons branch off to synapse with 4 more nuclei (LMN's): CN-IX, CN-X, CN-XI, CN-XII

OVERVIEW: all these nuclei contain neurons that allow swallowing, tongue-movement, and facial/jaw muscles

RULES: Neurons to the lower half of face have contralateral control, and those to upper face have bilateral control.

You've got this!

Dive deeper and explain the pathways for the "Rubrospinal" extra-pyramidal tract (in the context of uploaded diagram)

Dive deeper and explain the pathways for the "Corticobulbar" pyramidal tract (in the context of uploaded diagram)

Dive deeper and explain the pathways for the "Corticospinal" pyramidal tract (in the context of uploaded diagram)

Dive deeper and explain the pathways for the "Vestibulospinal" extra-pyramidal tract (in the context of uploaded diagram)

Don't know?

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Term 🔊

Table to help memorize the "Motor Tracts"

You're doing great!

Don't know?

Term 🔊

Outline the "extra-pyramidal" motor tracts of the CNS

Nice work!

1. Rubrospinal (fine motor control)
2. Reticulospinal
 - Medial reticulospinal (contraction increases tone)
 - Lateral reticulospinal (relaxation decreases tone)
3. Vestibulospinal (balance and posture)
4. Tectospinal/Colliculospinal (head co-ordination)

1. Autonomic Control
 - Cardiac = sympathetic and parasympathetic functioning
 - Respiratory = chemoreceptors (detect blood-oxygen concentration)
 - Reflexes = vomiting, coughing, sneezing, swallowing
 - Vasomotor = baroreceptors (detect blood-pressure changes)
2. Decussation of sensory fibers
 - Obex

1. Corticospinal tract (voluntary control of body)
 - Lateral corticospinal
 - Anterior corticospinal
2. Corticobulbar tract (voluntary control of face, head, and neck)

1. Autonomic nerve pathways
2. Integration of information
3. **Exerts some control over:**

- motor roles: eye movement, facial expressions, chewing, swallowing, secretion of tears/saliva

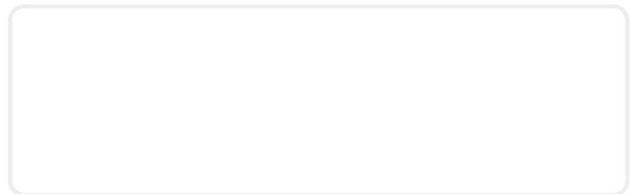
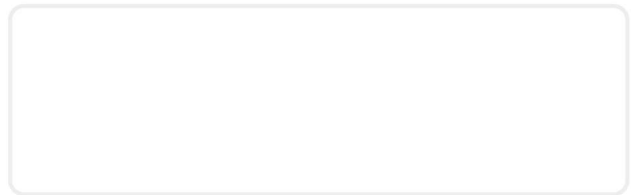
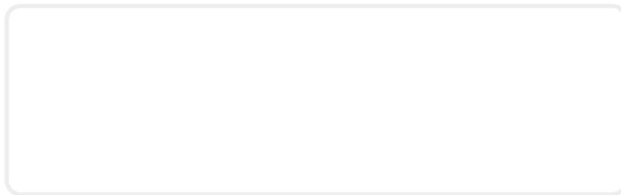
Don't know?

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Term 🔊

Diagram of a cross-section for the Pons

Awesome!



Don't know?

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Term 🔊

Dive deeper and explain the pathways for the "Tectospinal" extra-pyramidal tract (in the context of uploaded diagram)

Excellent!

- Originates in the Red Nucleus in the midbrain
- Continues on through the Medial lemniscus (in the midbrain) and the Pons and then branches-off at a LMN in the medulla.
- The tract continues on through the Lateral Reticular Nucleus in the medulla and branches-off at a LMN in the Cervical cord before ending at LMN in the Lumbar cord

OVERVIEW: all these nuclei contain neurons that allow for involuntary arm-movements (for balance) and body support

- Originates in the Superior Colliculus in the midbrain
- Continues on through the Pons and the Medulla, ending at medial LMN's in the Cervical cord

OVERVIEW: all these nuclei contain neurons that allow for involuntary head-movements

- Originates in the either the Medial Vestibular Nucleus or the Lateral Vestibular Nucleus in the Pons-Medulla junction
- MVN tract continues on through the upper medulla and branches off at the lower medulla with 1 of tract's neurons terminating as an input here. The 2nd neuron of this tract continues on through the lower medulla to LMN in the C-III of the Cervical cord
- LVN tract continues on through the upper medulla and lower medulla and terminates/branches-off at a LMN at each of the following structures: C-III of the Cervical cord, T-IV of Thoracic cord, L-III of the Lumbar cord, and S-II of the Sacral cord

OVERVIEW: all these nuclei contain neurons that allow for posture and balance

- Originates in either the Pontine Reticular Formation or the Medullary Reticular Formation
- Pontine tract continues on through the Medulla and ends at the LMN's in the C-VIII of Cervical cord
- Medullary tract continues from its origin to its endpoint at the LMN's in the C-VIII of Cervical cord

OVERVIEW: all these nuclei contain neurons that allow for various involuntary motor-movements, and assists with balance

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Definition

- Originates in the Red Nucleus in the midbrain
- Continues on through the Medial lemniscus (in the midbrain) and the Pons and then branches-off at a LMN in the medulla.
- The tract continues on through the Lateral Reticular Nucleus in the medulla and branches-off at a LMN in the Cervical cord before ending at LMN in the Lumbar cord

OVERVIEW: all these nuclei contain neurons that allow for involuntary arm-movements (for balance) and body support

You've got this!

Dive deeper and explain the pathways for the "Vestibulospinal" extra pyramidal tract (in the context

Dive deeper and explain the pathways for the "Reticulospinal" extra pyramidal tract (in the context

Dive deeper and explain the pathways for the "Rubrospinal" extra-pyramidal tract (in the context of uploaded diagram)

Dive deeper and explain the pathways for the "Tectospinal" extra-pyramidal tract (in the context of uploaded diagram)

Don't know?

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Term 🔊

What is the "Trigeminal nerve?"

You're doing great!

stretch and chemoreceptors of cardiovascular, respiratory, and intestinal tracts

cranial nerves IX and X:
-speech
-swallowing

cranial motor nerve nucleus

cranial nerve V:
-face sensations
-biting and chewing motor functions

Don't know?

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- Originates in either the Pontine Reticular Formation or the Medullary Reticular Formation
- Pontine tract continues on through the Medulla and ends at the LMN's in the C-VIII of Cervical cord
- Medullary tract continues from its origin to its endpoint at the LMN's in the C-VIII of Cervical cord

OVERVIEW: all these nuclei contain neurons that allow for various involuntary motor-movements, and assists with balance

Nice work!

Dive deeper and explain the pathways for the "Vestibulospinal" extra-pyramidal tract (in the context of uploaded diagram)

Dive deeper and explain the pathways for the "Tectospinal" extra-pyramidal tract (in the context of uploaded diagram)

Dive deeper and explain the pathways for the "Rubrospinal" extra-pyramidal tract (in the context of uploaded diagram)

Dive deeper and explain the pathways for the "Reticulospinal" extra-pyramidal tract (in the context of uploaded diagram)

Don't know?

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Term 🔊

What are the descending motor-tracts and pathways of the human brainstem and spinal cord?

Awesome!

- Originates in either the Pontine Reticular Formation or the Medullary Reticular Formation
- Pontine tract continues on through the Medulla and ends at the LMN's in the C-VIII of Cervical cord
- Medullary tract continues from its origin to its endpoint at the LMN's in the C-VIII of Cervical cord

OVERVIEW: all these nuclei contain neurons that allow for various involuntary motor-movements, and assists with balance

- Originates in the Red Nucleus in the midbrain
- Continues on through the Medial lemniscus (in the midbrain) and the Pons and then branches-off at a LMN in the medulla.
- The tract continues on through the Lateral Reticular Nucleus in the medulla and branches-off at a LMN in the Cervical cord before ending at LMN in the Lumbar cord

OVERVIEW: all these nuclei contain neurons that allow for involuntary arm-movements (for balance) and body support

- Originates in the Superior Colliculus in the midbrain
- Continues on through the Pons and the Medulla, ending at medial LMN's in the Cervical cord

OVERVIEW: all these nuclei contain neurons that allow for involuntary head-movements

1. Corticospinal tracts
 - lateral & anterior
2. Rubrospinal tract
3. Reticulospinal tracts
 - lateral & medial
4. Olivospinal tract
5. Vestibulospinal tract
6. (Tecto/Colliculo) spinal tract

Don't know?

20 of 20

Term 🔊

What is the "Solitary tract?"

Excellent!

pressure and temperature receptors
of the skin and muscles

pain and touch receptors of the
peripheral nerves

visual and auditory receptors of the
cranial nerves

stretch and chemoreceptors of
cardiovascular, respiratory, and
intestinal tracts

Don't know?